

DAY 3 Milestone:

OVERVIEW :

Live Data Simulation & Model Integration

Simulate real-time rainfall, water level, and pressure data and transmit them to cloud storage.

Integrate live data with the trained ML model to generate safety predictions.

Components Used :

- Rainfall Data Simulator
- Water Level Sensor Simulation
- Pressure Sensor Simulation
- Cloud Storage
- Trained ML Models:
 - XGBoost
 - LightGBM
 - LSTM
- Prediction Pipeline

System Workflow :

1. Generate real-time synthetic data for:
 - Rainfall intensity
 - Reservoir water level
 - Dam pressure
2. Add timestamp to each data entry
3. Transmit simulated data to cloud storage in real-time
4. Pre-process incoming data:
 - Handle missing values
 - Normalize features
5. Feed processed data into ML models

6. Generate outputs:

- Predicted water level
- Risk classification (Low / Medium / High)

Data Simulation Process :

The real-time data is simulated using random and pattern-based generation techniques to reflect realistic dam conditions:

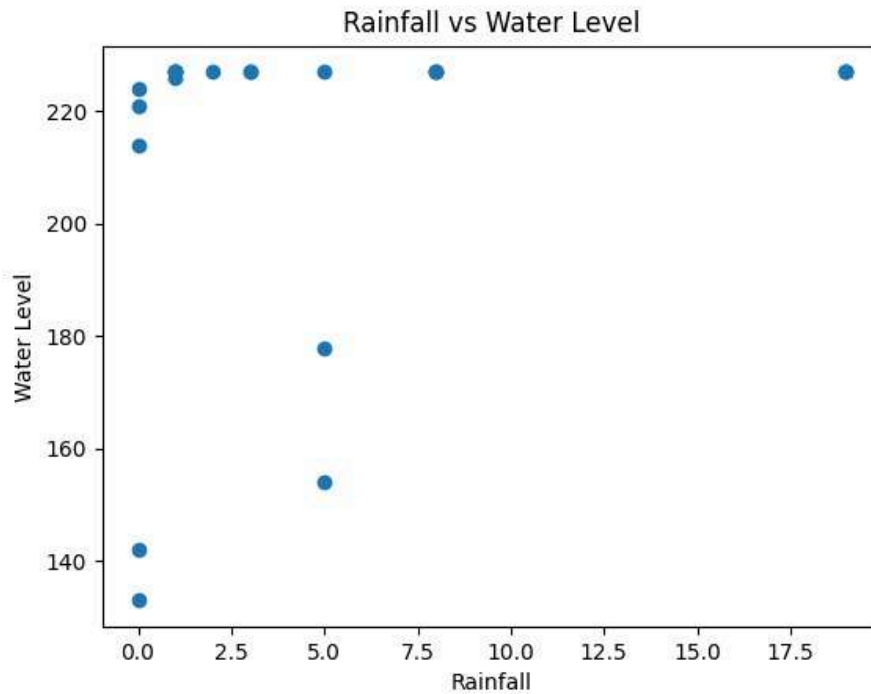
- Light rainfall → small increase in water level
- Heavy rainfall → rapid increase in water level and pressure
- Pressure increases with water level
- The simulation mimics real-world dam behavior under varying rainfall conditions

The simulation runs continuously to imitate live sensor streaming.

Real-Time Data Simulation :

Rainfall vs Water Level

- Shows system behavior
- Shows simulation logic



Model Integration :

The trained ML models are integrated into the pipeline to process incoming real-time data.

Models Used:

- **XGBoost** → Fast and accurate predictions
- **LightGBM** → Efficient with large data
- **LSTM** → Handles time-series forecasting

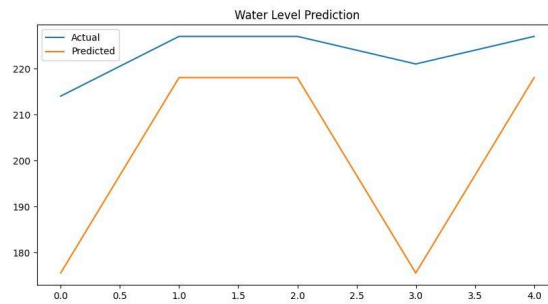
Functionality:

- Predict next water level
- Classify safety risk level

Model Prediction :

Water Level Prediction (Actual vs Predicted)

- Shows ML model working



Output Generated :

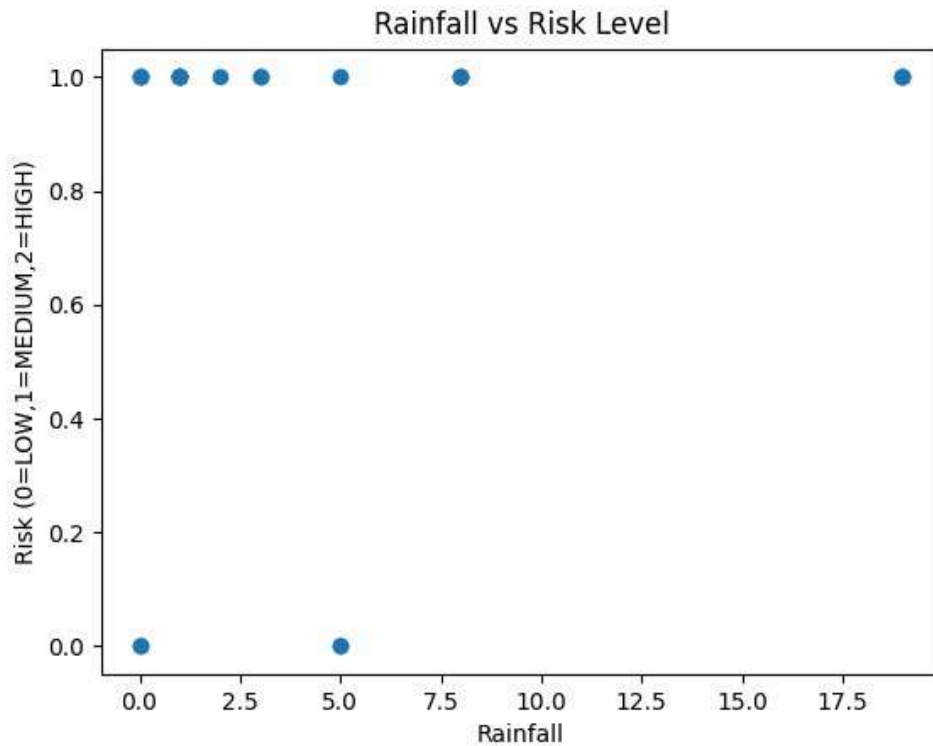
The system produces:

- Predicted Water Level
- Risk Level (Low / Medium / High)
- Timestamped Predictions

Prediction Results & Risk Classification Output :

Rainfall vs Risk Level

- Shows classification
- Directly related to your project



Sample Output Table :

Timestamp	Rainfall	Water Level	Pressure	Predicted Level	Risk
10:00 AM	5 mm	120 m	30 psi	122 m	Low
10:05 AM	20 mm	130 m	40 psi	138 m	Medium
10:10 AM	50 mm	150 m	60 psi	165 m	High

Conclusion :

This milestone successfully demonstrates the transition from static data to a real-time monitoring system. By simulating live sensor data and integrating it with trained ML models, the system is able to generate continuous predictions for water levels and classify risk conditions. This enables proactive decision-making and improves dam safety by identifying potential risks at an early stage.